

function testing on all units to be witnessed. Test procedure and program shall be submitted for approval prior to test conduction. Allow for test expenses including business class transportation and hotel accommodation as necessary for two persons from the Engineer's/ Employer's side.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Approved Manufacturers: Subject to compliance with requirements, provide products by one of the following or approved equal:

1. PETROKIMA (SIEMENS COMPONENTS) (EGYPT)
2. Schneider Electric (EGYPT)
3. Siemens /EMAS (Egypt)
4. ABB OR CERTIFIED PANEL BUILDER LIKE: (Egypt)
 - a. GMS
 - b. SMART INDUSTRIES
 - c. SYNCRO MISR
 - d. LECTRO

Schneider Electric

2.2 VARIABLE FREQUENCY DRIVE

- A. Furnish complete variable frequency drives as specified herein for the fans and pumps designated to be variable speed. All standards and optional features shall be included within the VFD enclosure, unless otherwise specified. VFD enclosure shall be **IP 21** for indoor and **IP 54 for outdoor** to IEC 60529 with inlet air filters. Drive shall be mounted on a floor stand.

Comply, IP55 conforming to IEC 60529, Refer to datasheet

- B. VFD shall be provided for motor starting and speed control where specified or shown on electrical or mechanical drawings.

Comply

- C. VFD shall convert three-phase, [50] hertz utility power to adjustable voltage and frequency, three phase power for stepless motor speed control from 10% to 100% of the motor's 50 hertz speed. Input voltage shall be as specified on the drawing schedules.

Comply

- D. VFD shall include a converter and an inverter section. The converter section shall convert fixed frequency and voltage AC utility power to DC voltage. All VFDs shall include input line reactors or an isolation transformer as required. The inverter section of the VFD shall invert the DC voltage into a quality output waveform, with adjustable voltage and frequency for stepless motor speed control. The VFD shall maintain a constant V/Hz ratio. The VFD and options shall be tested to IEC 60947-4-1.

AC Line Reactor added

- E. Power line noise shall be limited to a voltage distortion factor and line notch as defined in IEC 61000 as follows:

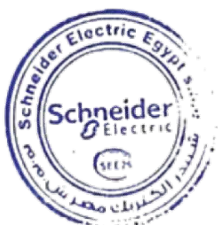
1. For motors with full load current below 16 A, comply with IEC 61000-3-2, and as a minimum use 6-pulse rectifier with DC buffer.
2. For motors with full load current below 75 A, comply with IEC 61000-3-4, and as a minimum use 6-pulse rectifier with 3% AC line reactor and DC buffer.
3. For motors with full load current exceeding 75 A and up to 450 A, comply with IEC 61000-3-4, and as a minimum use 6-pulse rectifier with 5% AC line reactor and DC buffer.

Comply, refer to EU declaration of Conformity

Comply with 61000-3-12, Refer to Catalog page 1/13



4. For motors with full load current exceeding 450 A, comply with IEC 61000-3-4, and as a minimum use 12-pulse rectifier with two secondaries galvanic isolating transformer to provide the necessary phase shift and 5% AC line reactor and DC buffer; as an alternative, active front end technique with LCL filter may be used.
- F. The VFD shall comply with EMC directive and standard IEC 61800-3 category C2 (1st environment), the VFD shall include EMC filter. **Comply, refer to catalog page 2/38**
- G. Motor noise as a result of the VFD shall be limited to 3 dB over across the line operation measured at 1 meter from the motor's centerline. **Comply with internal VFD Setting, Refer to Programming Manual Page 191**
- H. VFD's full load amp rating shall meet or exceed NFPA 70, NEC Table 430-250. **Comply, Refer to catalog page 2/7**
- I. Protective Features shall include the following:
1. Individual motor overload protection for each motor controlled. **Comply, refer to datasheet**
 2. Protection against input power under-voltage (for non-emergency motors only), over-voltage, and phase loss. **Comply, refer to datasheet**
 3. Protection against output current overload and instantaneous over-current. **Comply, refer to datasheet**
 4. Protection against over-temperature within the VFD enclosure. **Comply, refer to datasheet**
 5. Protection against over-voltage on the DC bus. **Comply, refer to datasheet**
 6. Thermistor motor protection relay interface. **Comply, refer to datasheet**
 7. Protect VFD from sustained power of phase loss. Under-voltage trip activates automatically when line voltage drops more than 10 percent below rated input voltage. **Comply, can be adjusted, Refer to programming Manual Page 560**
 8. Automatically reset faults due to under-voltage, over-voltage, phase loss, or over-temperature. **Comply, refer to programming Manual Page 532**
 9. Protection against output short circuit and motor winding shorting to case faults. **Comply, refer to datasheet**
 10. Status lights or digital display for indication of individual fault conditions. **Comply, refer to catalog page 2/14**
 11. Controller capable of operating without a motor or any other equipment connected to the drive output to facilitate start-up and troubleshooting. **Comply, refer to programming manual page 21 and 22**
 12. Input line reactors shall be provided to minimize harmonics introduced to the AC line and to provide additional protection to AC line transients. **Comply, refer to Catalog Page 2/41**
- J. Interface Features for the VFD will include:
1. Door mounted Hand/Off/Auto selector switch to start and stop the VFD. In the Auto position, the VFD will start/stop from the controller. In the Hand position, the VFD will run regardless of the remote contact position. **Comply from VFD screen**
 2. Manual speed control capability. **Comply, refer to installation manual page 225**
 3. Local/Remote selector switch. In the Remote position, motor speed is determined by the follower signal. In the local position, motor speed is determined by the manual speed control. **Comply**
 4. Power/On light to indicate that the VFD has tripped on a fault condition. **Comply, refer to catalog page 2/14**
 5. Digital meter with selector switch to indicate percent speed and percent load. **Comply, refer to catalog page 2/14**
 6. A set of form-C dry contacts, to indicate when the VFD is in the Run mode. **Comply, Installation Manual Page 222**
 7. A set of form-C dry contacts, to indicate when the VFD is in the Fault mode. **Comply, Installation Manual Page 222**
 8. A 0 to 10 V output signal to vary in direct proportion to the controller's speed. **Comply, Installation Manual Page 225**
 9. VFD to have terminal strip and N.C. safety contacts such as freezstats, smoke alarms, etc. VFD to safety shut down in drive or bypass mode when contacts open. **Comply, Installation Manual Page 224**
- K. Adjustments will include the following:
1. Maximum speed, adjustable 15 to 100 percent base speed. **Comply, Refer to Programming Manual Page 56**
 2. Acceleration time, adjustable 3 to 60 seconds. **Comply, Refer to Programming Manual Page 357**



3. Deceleration time, adjustable 3 to 60 seconds with override circuit to prevent nuisance trips if decel time is set too short. Comply, Refer to Programming Manual Page 357
4. Current limit, adjustable 0 to 105 percent. Comply, Refer to Programming Manual Page 172
5. Overload trip set point. Comply, Refer to Programming Manual Page 357
6. Offset and gain to condition the input speed signal Comply, Refer to Programming Manual Page 305 and 306

L. Service conditions will cover:

1. Ambient temperature, 0 to 50 deg C. Comply, Refer to datasheet
2. 0 to 95 percent relative humidity, non-condensing. Comply, Refer to datasheet
3. Elevation to 1,000 meters without derating. Comply, Refer to datasheet
4. AC line voltage variation, -10 to +10 percent of nominal. Comply, Refer to datasheet

M. Control interface: VFD shall be equipped with field-bus module for remote control, monitor and display VFD status and alarms from the BMS. Coordinate with BMS for the interface protocol to be adapted and provide the necessary adapter modules accordingly.

Comply, Refer to Catalog Page 1/6

Integrated: Modbus/TCP, Modbus serial link

Optional: Ethernet/IP, Modbus TCP and MD-Link dual port, CANopen RJ45 daisy chain, SUB-D, and screw terminal block, PROFINET, PROFIBUS DP V1, DeviceNet, BACnet MS/TP, POWERLINK

2.3 STARTERS

- A. Starters for three phase motors to be magnetic type to automatically disconnect motor from power supply in case of supply failure, excessive voltage drop, overcurrent and lack of balance in phases. Overload trips to be provided for three phases.
- B. Motor Data: Obtain from equipment supplier before ordering any motor starter, or check motor nameplate for full load current rating and allowable temperature rise in order to select proper overload thermal element for motor starter.
- C. Short circuit protection device fitted to starter to be independent of controller and overload protection.
- D. Control voltage for starters and control circuits shall not exceed 110 V.
- E. Step Down Control Circuit Transformers: Two winding isolating type.
- F. Control Circuit Protection: Use high rupturing capacity fuses or circuit breakers.
- G. Auxiliary supply for controls other than from main power circuit, to be effectively isolated by auxiliary contacts on main isolator.
- H. Control devices on starters to be as follows unless otherwise indicated or required by driven equipment: start stop push buttons, one red pilot light for "running", one green pilot light for "stopped", one amber pilot light for overload trip, and one reset push button.
- I. All motors shall be protected with overload protection class 10 to IEC 60947-4-1, unless otherwise required by the motor manufacturer.
- J. Starter type A for single phase motors not exceeding 1/2 HP to be surface or flush mounted, manual two pole toggle type, for non reversing across the line starting, fitted with one overload element.
- K. Starter type B for three phase motors not exceeding 10 HP to be direct on line, non reversing, magnetic type, with manual reset, 3 pole overload relay and under voltage protection (for non-emergency motors only), unless otherwise required by local regulations or shown on drawings.

